

LESSON 6.5

CONVERTING UNITS OF VOLUME



LESSON INTRODUCTION

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

LEARNING TARGETS

- I can convert liquid measurements and volume across different measurement systems.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths, and conversions within the same measurement system.

WORKING TOWARD

MA.8.AR.3 Extend understanding of proportional relationships to two-variable linear equations.

ESSENTIAL QUESTIONS

- Why is the volume of solid figures represented with cubic units?
- How do you convert volume between units of measure?

LESSON NARRATIVE

Students continue building on their conversion methods in various contexts, specifically targeting volume measurements in this lesson. Instruction and practice include finding volume with conversion rates rather than what students might know about length, width, and height.

COMPONENT SUMMARY

6.5.1 WARM-UP (~5 MIN.)

Currency conversion

6.5.3 TRY IT (5-10 MIN.)

Practice converting measures of liquid volume

6.5.5 GUIDED INSTRUCTION (5-10 MIN.)

Convert measurements of volume using cubic units

6.5.7 WRAP-UP (~5 MIN)

Identify correct conversion factors for liquid measurement and convert fluid ounces to liters

6.5.2 ACTIVITY (10 MIN.)

Video exploration for conversions and equivalence

6.5.4 GUIDED PRACTICE (5-10 MIN.)

Use conversion tables and rates to convert liquid volume between systems

6.5.6 YOUR TURN! (5-10 MIN.)

Practice converting between U.S. customary, metric, and cubic units

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- Cubic unit
- Liquid volume
- Solid volume

MATERIALS

- (Optional) Two-liter bottle, gallon bottle, water

ADDITIONAL PREPARATION

- 6.5.2 Activity requires either a video or an in-person demonstration. The video can be used whole group, small group, or as a self-paced activity. If technology is not an option, consider bringing in a two-liter bottle and a gallon of water to demonstrate live.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse the units for liquid volume and capacity with the units for weight and mass.
- Students may use the same multiplier for converting lengths instead of cubing the multiplier when converting volume.
- Students may not pay attention to the units of measurement and try to perform operations without first considering an appropriate conversion factor.

THINGS TO CONSIDER

- Provide opportunities for students to reflect on their methods and approaches to continue building fluency.
- Provide students with a calculator for performing calculations with the expectation that they write their calculations first.

LITERACY AND LANGUAGE ROUTINES

Stronger and Clearer

Provide scaffolding for students to apply the Stronger and Clearer routine during 6.5.2 Activity. This routine positions students in crafting more precise and mathematical responses to each prompt, leading to better fluency.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Fluency includes the ability to discern the most efficient methods in problem-solving. Students make decisions throughout this lesson in taking different approaches to the same answer. Build-in opportunities to prompt students to compare their process and not their product and share which were most efficient.

DIFFERENTIATE INSTRUCTION

6.5.2 Activity provides a variety of entry points for students who learn and make sense of information in a variety of ways. Consider bringing in household objects for students to use as manipulatives to enhance the conceptual understanding of the relationship in equivalence between different measures. For example, some measuring cups provide both metric and U.S. customary measures. Consider bringing in materials for students to actually demonstrate the video for a live demonstration for hands-on learning.

6.5.1 WARM-UP: MAYAN CURRENCY

ENRICHMENT

- Have you ever exchanged something other than money to buy or borrow something? How did you determine what it was worth?
- Why do you think coins made of gold or silver had more value than cacao beans?



6.5.1 Warm-Up: Mayan Currency

The ancient Maya used cacao beans – the main ingredient in chocolate – as a form of currency.



They did not use coins as money, but instead mostly bartered and traded items such as tobacco, maize (corn), and clothing. Archeologists have found that chocolate was used as currency by the 8th century CE, in the form of fermented and dried cacao beans. Maya artwork indicates that cacao beans could be exchanged for goods and services, including payment for labor and paying taxes.

Cacao, just like any other currency, could be converted to other kinds of money, including Spanish gold and silver coins such as the real. It has been found that 200 cacao beans were worth 1 Spanish real.

1. If 24,000 cacao beans were worth 15 pesos, what was the rate of cacao beans for 1 peso?

$$\frac{24,000 \text{ cacao beans}}{15 \text{ pesos}} = \frac{1600 \text{ cacao beans}}{1 \text{ peso}}$$

2. How many pesos were equal to 1 Spanish real?

$$\frac{1 \text{ real}}{1} \times \frac{200 \text{ cacao beans}}{1 \text{ real}} \times \frac{1 \text{ peso}}{1600 \text{ cacao beans}} = \frac{1}{8} \text{ pesos} \approx 0.125 \text{ pesos}$$



6.5.2 Activity: Pouring Liquids to Find Conversion Rates

Each of us uses containers that have volume. Most of those containers are labeled with both U.S. customary measurements and liquid metric measurements.

- 1. List some U.S. customary measurements and liquid metric measurements.
Answers will vary. Liters; Ounces; Cups; Teaspoons

Many soft drinks come in 2-liter bottles, while milk is often sold by the gallon.

2-Liter Soda Bottle	1 Gallon Milk Container
	

- 2. Which container do you think holds more liquid? Explain your reasoning.
Answers will vary.
The one-gallon milk container appears wider and will carry more liquid. The soda bottle is taller and will contain more liquid.

Both liters (L) and gallons (gal) are units used to measure liquid volume. Liters are units within the metric system, while gallons are part of the U.S. customary system.

6.5.2 ACTIVITY: POURING LIQUIDS TO FIND CONVERSION RATES

TEACHER MOVES

Stronger and Clearer

Questions 1-3 position students for more accurate or precise predictions in preparation for the video.

After students write their reasoning, prompt them to verbally share with a partner. After the partner listens, have the partner ask a question that starts with “What did you mean by ___?” Then the first partner responds, and both revise their written answers. Have students do this protocol at least twice to help them strengthen their mathematical reasoning, explicitly making connections between their response to question 2 and question 3 in their final response.

Questions 4-6 offer additional opportunities for students to build their math reasoning and make connections between verbal descriptions and algebraic representation.

TEACHER MOVES

The video could be viewed as a whole group after students have completed questions 1-3, or students can watch in small groups on devices and complete this activity at their own pace.

6.5.2 ACTIVITY: POURING LIQUIDS TO FIND CONVERSION RATES (CONTINUED)

ENRICHMENT

What are some of the features you saw or looked for that helped inform a more precise answer?

DIFFERENTIATE INSTRUCTION

Play the video twice if students need some more support outside of the audio and visual presentation. If possible, turn on captions for the video to support students who learn best by reading.

QUESTIONING

Prompt students to create their own video for homework and project-based learning.

3. What do you predict is the relationship between liters and gallons based on the size of the containers? Write your prediction under the initial prediction heading in the table below.

Answers will vary.

Initial Prediction	Revised Prediction
<u>1</u> gallon(s) $\approx$ <u>2</u> liter(s)	<u>1</u> gallon(s) $\approx$ <u>2.5</u> liter(s)
<u>1</u> gallon(s) $\approx$ <u>1</u> liter(s)	<u>1</u> gallon(s) $\approx$ <u>2</u> liter(s)

Watch the short video of liquid from one container being poured into the other.



4. Based on your initial prediction and the amount of liquid in each container now, discuss the relationship between liters and gallons. Then, write a revised prediction in the table above.

Answers will vary.

Continue watching the remainder of the video.

5. Based on the video, what is your final estimate for the relationship between liters and gallons?

Answers will vary.

Final Estimate
<u>1</u> gallon(s) $\approx$ <u>3</u> liter(s)
<u>1</u> gallon(s) $\approx$ <u>4</u> liter(s)

6. How close is your estimate to the actual conversion rate?

The big reveal ...



Actual Conversion Rate
<u>1</u> gallon(s) $\approx$ <u>3.785</u> liter(s)



6.5.3 Try It: Converting Liquid Volume Measurements

1. Aleah is mixing concrete for a home-improvement project. According to the bag directions, she needs 6 gallons (gal) of water to mix the amount of concrete needed. The mixing bucket only has markings in liters (L). How many liters of water does she need to measure out for mixing the concrete? Round to the nearest tenth.

$$6 \text{ gal} \times \frac{3.785 \text{ L}}{1 \text{ gal}} = 22.7 \text{ L}$$

2. Most cars have a fuel tank that can hold between 45 and 65 liters (L) of gasoline. What is the range of gasoline a typical fuel tank can hold in terms of gallons (gal)? Round to the nearest hundredth.

$$11.89 - 17.17 \text{ gal}$$

$$45 \text{ L} \times \frac{1 \text{ gal}}{3.785 \text{ L}} \approx 11.89 \text{ gal}$$

$$65 \text{ L} \times \frac{1 \text{ gal}}{3.785 \text{ L}} \approx 17.17 \text{ gal}$$

6.5.3 TRY IT: CONVERTING LIQUID VOLUME MEASUREMENTS

TEACHER MOVES

MA.K12.MTR.3.1: Math Fluency

Consider having students turn and talk to discuss the different approaches in questions 1 and 2 to build fluency. Use the following prompts: • What is different between how we used the conversion rates in questions 1 and 2?

- Are these the only ways we could have completed these conversions?
- Are there other ratios that we could have used? Explain which makes the most sense in this context.



6.5.4 Guided Practice: Converting Liquid Volume Measurements

The units commonly used for liquid capacity within each system are shown, listed from smallest to largest.

U.S. Customary System	Metric System
fluid ounces (fl oz)	
cups (c)	milliliters (mL)
pints (p)	liters (L)
quarts (qt)	
gallons (gal)	

The tables below show the conversion rates for common units of liquid capacity in the U.S. customary and metric systems.

U.S. Customary System to Metric System	
1 fluid ounce (fl oz)	= 28.413 milliliters (mL)
1 cup (c)	= 236.588 milliliters (mL)
1 fluid ounce (fl oz)	= 0.030 liters (L)
1 cup (c)	= 0.24 liters (L)
1 quart (qt)	= 0.946 liters (L)
1 gallon (gal)	= 3.785 liters (L)
Metric System to U.S. Customary System	
1 milliliter (mL)	= 0.034 fluid ounces (fl oz)
1 milliliter (mL)	= 0.004 cups (c)
1 liter (L)	= 33.814 fluid ounces (fl oz)
1 liter (L)	= 4.23 cups (c)
1 liter (L)	= 1.057 quarts (qt)
1 liter (L)	= 0.264 gallons (gal)

Student answers may vary for each conversion depending on which conversion rate they use, customary to metric or metric to customary.

1. Use the conversion rates to find each value below. Round to the nearest hundredth.
 - a. 240 mL $\approx$ 1.01 c $240 \text{ mL} \times \frac{1 \text{ c}}{236.588 \text{ mL}}$
 - b. 100 fl oz $\approx$ 2.96 L $100 \text{ fl oz} \times \frac{1 \text{ L}}{33.814 \text{ fl oz}}$
2. There are 100 centiliters (cL) in a liter. How many centiliters are in 2.75 quarts? Round to the nearest hundredth.

$$2.75 \text{ qt} \times \frac{0.946 \text{ L}}{1 \text{ qt}} \times \frac{100 \text{ cL}}{1 \text{ L}} \approx 260.15 \text{ cL}$$

6.5.4 GUIDED PRACTICE: CONVERTING LIQUID VOLUME MEASUREMENTS

DIFFERENTIATE INSTRUCTION

Consider bringing in objects and measuring tools that depict both measuring systems as anchors and manipulatives for students.

6.5.5 GUIDED INSTRUCTION: CUBIC UNITS OF VOLUME

TEACHER MOVES

MA.K12.MTR.3.1: Math Fluency

Consider prompting students to discuss and surface which method is more efficient.

ENRICHMENT

Consider asking students to explain which method they used.



6.5.5 Guided Instruction: Cubic Units of Volume

Volume of solid figures is measured in cubic units. In some settings, liquid volume is also measured in cubic units. For instance, in many medical situations, medicines and other fluids are measured as cubic centimeters.

1 cubic centimeter = 1 milliliter

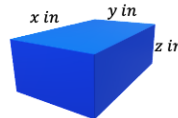
- Complete the equation. Show your work and round to the nearest hundredth.

4.5 fl oz $\approx$ 127.86 cubic centimeters (cm^3)

$$4.5 \text{ fl. oz.} \times \frac{28.413 \text{ mL}}{1 \text{ fl. oz.}} \times \frac{1 \text{ cm}^3}{1 \text{ mL}}$$

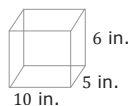
For solid volume, the conversion rates can be determined using the conversion rates for length.

There are two ways to find the volume of a rectangular prism in cubic centimeters when the prism's side measurements are given in inches.



Method 1	Method 2
Convert each side length to centimeters and then calculate the volume of the figure in cubic centimeters.	Calculate the volume of the figure in cubic inches and then convert the volume to cubic centimeters.

- Work with your partner to determine a conversion factor that can be used to convert cubic inches (in^3) to cubic centimeters (cm^3). Round to the nearest thousandth. Use the known conversion of 1 inch (in.) = 2.54 centimeters (cm). $(1 \text{ in.})^3 = (2.54 \text{ cm})^3$
 $1 \text{ cubic inch (in.}^3) \approx 16.387 \text{ cubic centimeters (cm}^3)$
- How many cubic centimeters can the rectangular prism with dimensions of 10 in. by 5 in. by 6 in. hold?



$$10 \times 5 \times 6 = 300 \text{ in}^3$$

$$300 \text{ in}^3 \times \frac{16.387 \text{ cm}^3}{1 \text{ in}^3} =$$

$$4916.1 \text{ cm}^3$$

6.5.6 YOUR TURN!

TEACHER MOVES

Notice the calculations in question 1 compared to the context and problem-solving in question 2.

QUESTIONING

For students who finish early, challenge them to choose one question and find at least one other method to find the answer and compare to determine which method they prefer.



6.5.6 Your Turn!

- Use conversion rates to find each value. Round to the nearest thousandth.
 - 6 cups $\approx$ 1419.528 milliliters
 $6 \text{ c} \times \frac{236.588 \text{ mL}}{1 \text{ c}}$
 - 300 milliliters $\approx$ 10.144 fluid ounces
 $300 \text{ mL} \times \frac{1 \text{ fl. oz.}}{29.574 \text{ mL}}$
 - 2 liters $\approx$ 2.114 quarts
 $2 \text{ L} \times \frac{1.057 \text{ qt.}}{1 \text{ L}}$
 - 10 gallons $\approx$ 37.854 liters
 $10 \text{ gal} \times \frac{3.785 \text{ L}}{1 \text{ gal}}$
- Mateo buys a package with 12 bottles of water. Each bottle holds 500 milliliters of water. How many fluid ounces are in the package?

$$500 \text{ mL} \times 12 = 6000 \text{ mL}$$

$$6000 \text{ mL} \times \frac{0.034 \text{ fl. oz.}}{1 \text{ mL}} = 204 \text{ fl. oz.}$$



6.5.7 Wrap-Up: Ticket Out the Door

1. For each pair of liquid measurements, circle the conversion factor to use to convert from the measurement on the left to the measurement on the right.

Conversion Table			
Converting from...			...to
fluid ounces	$\frac{1 \text{ fl oz}}{28.413 \text{ mL}}$	$\frac{28.413 \text{ mL}}{1 \text{ fl oz}}$	milliliters
milliliters	$\frac{0.004 \text{ c}}{1 \text{ mL}}$	$\frac{283.588 \text{ mL}}{1 \text{ c}}$	cups
liters	$\frac{1.057 \text{ qt}}{1 \text{ L}}$	$\frac{1 \text{ L}}{1.057 \text{ qt}}$	quarts
gallons	$\frac{1 \text{ gal.}}{3.785 \text{ L}}$	$\frac{1 \text{ L}}{0.264 \text{ gal.}}$	liters

2. Carmita is mixing sweet tea and lemonade to make her favorite beverage. She mixes 20 fluid ounces (fl oz) of sweet tea with 16 fluid ounces of lemonade. How many liters (L) of the beverage did she mix?

$20 + 16 = 36 \text{ fl. oz.} \times \frac{0.030 \text{ L}}{1 \text{ fl. oz.}} = 1.08 \text{ L}$

6.5.7 WRAP-UP: TICKET OUT THE DOOR

TEACHER MOVES

In 1983, a Boeing 767 ran out of fuel midflight because of a conversion error. (Luckily, the pilots were able to land the plane successfully.) Challenge students to describe a scenario for which converting measurements correctly is crucial.